



American International University-Bangladesh (AIUB)

Department of Computer Science

Faculty of Science & Technology (FST)

Disaster Relief Coordination and Resource Management System

Semester: Summer 25-26

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1. PROJECT PROPOSAL

1.1 Background to the Problem

Bangladesh is highly vulnerable to natural disasters such as floods, cyclones, river erosion, landslides, and thunderstorms. These disasters often cause loss of life, damage to property and infrastructure, displacement of people, and shortages of basic necessities such as food, clean water, medicine, and shelter. Therefore, quick and effective disaster response is important for protecting affected communities.

During disasters, different organizations such as government agencies, NGOs, volunteers, hospitals, and local authorities work together to provide emergency services and relief. However, coordination among these organizations can be difficult because information is often shared through different communication channels and records may be managed manually. This can lead to delayed information, poor communication, duplicate relief distribution, and inefficient use of available resources.

Affected people may also face difficulties in reporting emergencies, requesting assistance, finding nearby shelters, and receiving reliable information. At the same time, relief organizations may have difficulty tracking available resources, managing volunteers, and monitoring relief activities.

Therefore, there is a need for a centralized digital system that can improve communication, coordinate relief activities, and manage resources more efficiently. The **Disaster Relief Coordination and Resource Management System** is proposed to address these challenges by providing a common platform for citizens, volunteers, relief organizations, and authorities to coordinate disaster response and manage relief resources effectively.

1.2 Selection of Process Model

- **Selected Process Model: Scrum**

For the development of the **Disaster Relief Coordination and Resource Management System**, the **Scrum model** has been selected. Scrum is an Agile software development framework that divides the project into small development cycles called **Sprints**. Each Sprint focuses on developing, testing, and completing a specific set of system features. Feedback from each Sprint is used to improve the system in the following Sprint. **Scrum Process**

The Scrum process can be divided into three main phases:

1. Pre-game Phase

The **Pre-game phase** focuses on preparing the project before the main development starts. During this phase, the project requirements are identified and analyzed. The main system features are added to the **Product Backlog** and prioritized. The development team also prepares the initial project plan, architecture, and design.

For this project, activities in the Pre-game phase include:

- Identifying system requirements.
- Identifying stakeholders and users.
- Preparing the Product Backlog.
- Prioritizing system features.
- Preparing the initial system design and architecture.
- Planning the first Sprint.

2. Game Phase (Development)

The **Game Phase** is the main development phase of Scrum. The project is divided into several Sprints. In each Sprint, selected requirements are analyzed, designed, developed, tested, and reviewed.

For example, the development of this system can be divided as follows:

Sprint Main Features

Sprint 1 User registration, login, and role management

Sprint 2 Emergency reporting and relief requests

Sprint 3 Shelter management and shelter locator

Sprint 4 Volunteer registration and task assignment

Sprint 5 Resource inventory and resource distribution

Sprint 6 Notifications and organization coordination

Sprint 7 Admin dashboard, reports, integration, and final testing

At the end of each Sprint, a working part of the system is produced and reviewed.

3. Post-game Phase

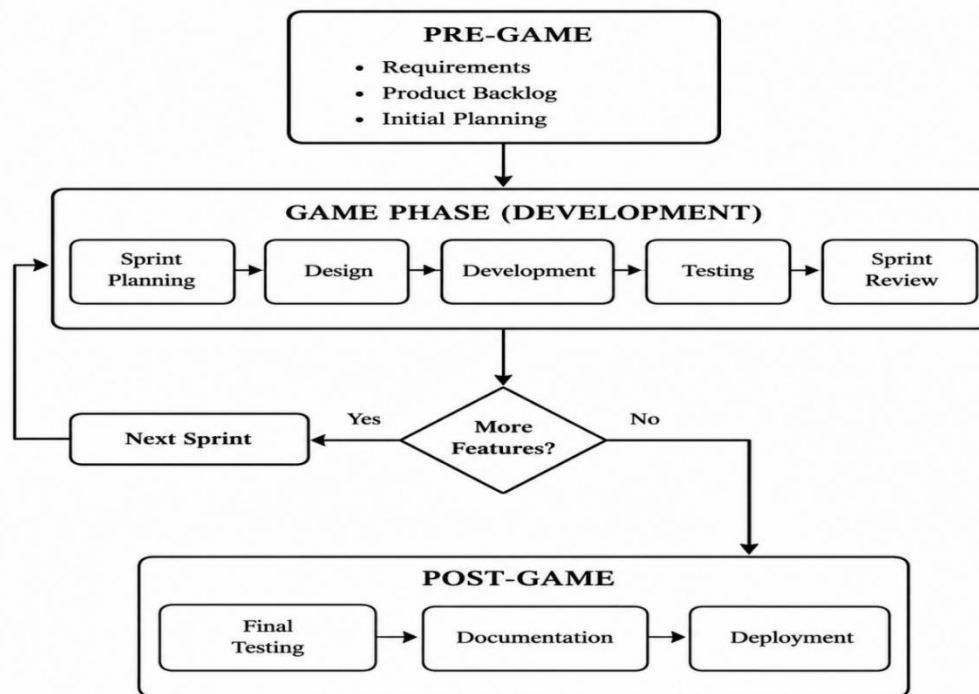
The **Post-game phase** takes place after the required features have been developed. The complete system is integrated and undergoes final testing. Documentation is completed, remaining issues are fixed, and the system is prepared for deployment.

Main activities include:

- Final system integration.
- System and acceptance testing.
- Bug fixing.
- Final documentation.
- System deployment.
- Final project evaluation.
- **Scrum Process Visualization**

- **Scrum Process Visualization**

The Scrum process for the proposed system can be represented as follows:



Scrum Roles and Responsibilities

Role	Responsibilities
Scrum Master	Ensures that Scrum practices are properly followed, facilitates Scrum meetings, helps the team remove obstacles, and supports effective communication among team members.
Product Owner	Identifies and prioritizes system requirements, manages the Product Backlog, represents stakeholder needs, and ensures that the developed system provides value to users.
Scrum Team	Analyzes requirements, designs, develops, tests, integrates, and delivers system features during each Sprint. The team works together to produce a working product increment.
Customer	Provides requirements, feedback, and suggestions about the system. For this project, customers can include disaster relief organizations, NGOs, government authorities, volunteers, and affected citizens.
Management	Provides project support, resources, budget, infrastructure, and overall organizational guidance. Management also monitors project progress and ensures that the project follows organizational goals.

For this project, stakeholders such as **citizens, volunteers, NGOs, government authorities, and relief coordinators** can provide feedback and requirements to the Product Owner.

- **Scrum Practices**

Scrum Practice	Description
Product Backlog & Sprint	The Product Backlog contains a prioritized list of system requirements and features. A Sprint is a short development cycle in which selected backlog items are developed and tested.
Effort Estimation & Sprint Backlog	The Scrum Team estimates the effort required for each backlog item. Selected items are then moved into the Sprint Backlog for development during the current Sprint.
Sprint Planning Meeting	The Product Owner and Scrum Team discuss and select the requirements to be completed in the upcoming Sprint. The team also plans how the selected work will be completed.
Daily Scrum Meeting	A short daily meeting where team members discuss their progress, planned work, and any problems or obstacles affecting development.
Sprint Review Meeting	At the end of the Sprint, the Scrum Team demonstrates the completed features to the Product Owner and customers/stakeholders. Feedback is collected for future improvements.
Sprint Retrospective	The Scrum Team reviews the completed Sprint, discusses what went well and what could be improved, and identifies improvements for the next Sprint.

Product Increment	A working and tested part of the Disaster Relief Coordination and Resource Management System produced at the end of a Sprint.
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The following Scrum practices will be used during the development of the proposed system:

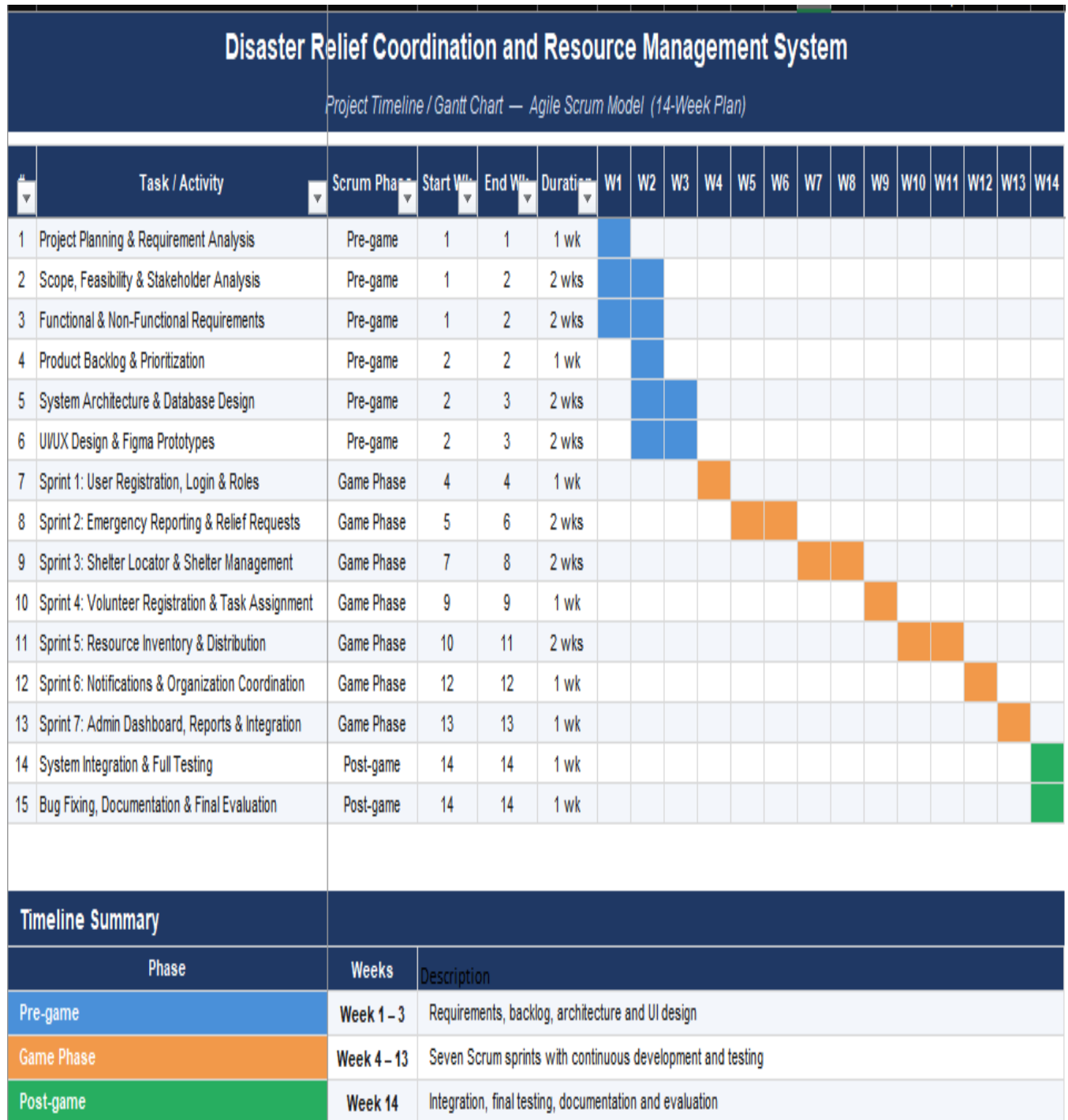
- **Why Scrum Was Chosen**

Scrum was selected for the **Disaster Relief Coordination and Resource Management System** because the project contains multiple modules that can be developed incrementally. Features such as emergency reporting, shelter management, volunteer coordination, resource management, notifications, and reporting can be completed through separate Sprints.

The main reasons for choosing Scrum are:

1. **Incremental Development:** The system can be developed feature by feature instead of developing the entire system at once.
2. **Flexible Requirements:** Requirements can be changed or added to the Product Backlog when new needs are identified.
3. **Continuous Testing:** Each feature can be tested during its Sprint, helping to identify and fix errors early.
4. **Regular Feedback:** Stakeholders can review completed features and provide feedback after each Sprint.
5. **Easy Project Management:** Dividing the project into smaller Sprints makes development easier to plan, monitor, and manage.
6. **Early Working System:** A working version of the system can be produced before the entire project is completed.
7. **Suitable for Multiple Users:** The system involves citizens, volunteers, NGOs, administrators, and relief coordinators. Scrum allows their requirements and feedback to be considered throughout development.

2. Timeline Chart / Gantt Chart



3. SOFTWARE REQUIREMENTS SPECIFICATIONS (SRS)

3.1 Scopes and Features

- **Scope of the Project**

Scope	Description
Citizen Emergency Assistance	Allows affected citizens to report emergencies and request rescue or relief assistance.
Shelter Management	Provides information about available shelters, locations, capacity, and current status.
Volunteer Coordination	Allows volunteers to register, provide their skills and availability, and receive relief tasks.
Resource Management	Helps relief organizations manage, track, allocate, and distribute resources such as food, water, medicine, and clothing.
Relief Organization Coordination	Allows NGOs, government agencies, and other organizations to coordinate disaster relief activities.
Emergency Communication	Supports communication and information sharing between citizens, volunteers, organizations, and administrators.
Notification Management	Provides disaster alerts, emergency updates, shelter information, and relief-related notifications.
Monitoring and Reporting	Allows authorized users to monitor relief activities and generate reports for decision-making.
User and Role Management	Manages different types of users and provides access based on their roles and permissions.

- Features of the Project

Features	Description
User Registration & Login	Allows citizens, volunteers, organizations, and administrators to create accounts and securely access the system.
Emergency Reporting	Enables citizens to report emergencies by providing details such as location, type of disaster, and required assistance.
Relief Request	Allows affected people to request specific assistance such as food, water, medicine, rescue, or shelter.
Emergency Request Tracking	Allows users and coordinators to check the status of submitted emergency or relief requests.
Shelter Locator	Helps users find nearby shelters and view their location, capacity, and availability.
Shelter Management	Allows authorized users to add, update, and monitor shelter information and occupancy.
Volunteer Registration	Allows people to register as volunteers and provide their skills, location, and availability.
Volunteer Task Assignment	Allows coordinators to assign rescue and relief tasks to suitable volunteers.
Resource Inventory Management	Tracks available relief supplies such as food, water, medicine, clothing, and emergency equipment.
Resource Allocation & Distribution	Helps organizations allocate resources to affected areas and record their distribution.
Organization Management	Allows NGOs and other relief organizations to manage their profiles and coordinate relief activities.
Real-Time Notifications	Sends alerts and updates about emergencies, relief requests, shelters, and assigned tasks.
Disaster Information Sharing	Provides important disaster-related information and updates to users.
Admin Dashboard	Provides administrators with an overview of users, requests, shelters, volunteers, resources, and relief activities.
Reports & Analytics	Generates reports on emergency requests, resource distribution, volunteer activities, and relief operations.
Role-Based Access Control	Provides different system permissions based on user roles such as citizen, volunteer, organization, and administrator.

3.2 User Story Table

As A/An	I want to	So that	Acceptance Criteria
Citizen	Create a new profile	I can enroll myself in the system	Account is created successfully and saved to the system (FR1.1)
	Login with my username and password	the system can authenticate me and I can trust it	After successful login, my dashboard and associated data are displayed (FR1.2, 1.3)
	View and update my profile information	my details stay accurate	Profile information list is shown, including an edit option, and updates are saved (FR1.4)
	Report a disaster-related emergency	relief authorities can be informed and respond quickly	Emergency type, description, location, and date/time are recorded (FR2.1, 2.2, 2.3)
	Request relief assistance such as food, water, medicine, clothing, shelter, or rescue	I can receive the help I need	Request is saved with type and required quantity/level of assistance (FR3.1, 3.2, 3.3)
	Track the status of my submitted emergency/relief requests	I know when help will arrive	Request status is displayed and updated as it changes (FR3.4, FR11.3)
	Search for nearby available shelters	I can find a safe place to stay	Shelter list is shown with location, capacity, and current availability (FR4.3)
	Receive notifications about my request status and disaster updates	I stay informed during the emergency	Notification is delivered to my account promptly after it is generated (FR9.2, 9.4)
Shelter Manager	Login with my username and password	the system can authenticate me and I can trust it	After successful login, the shelter manager dashboard and associated data are displayed (FR1.2, 1.3)
	Add and update shelter information such as name and location	citizens can find accurate shelter details	Shelter information is saved and displayed correctly (FR4.1, 4.2)

As A/An	I want to	So that	Acceptance Criteria
	Update shelter capacity and available space	citizens see accurate real-time availability	Updated capacity/availability is reflected immediately in the shelter list (FR4.4)
	View relief and shelter requests submitted by citizens	I can prepare and allocate space for the people coming to my shelter	Shelter-related requests are displayed and can be searched/filtered (FR3.1, FR13.2)
	Monitor the current status and occupancy of my shelter	I can manage capacity effectively during a disaster	Shelter status and occupancy details are displayed and kept up to date (FR4.2, FR4.4)
Volunteer	Register as a volunteer with my skills, location, and availability	I can be assigned relief tasks	Volunteer profile is saved and marked as pending verification (FR5.1, 5.2)
	Login with my username and password	the system can authenticate me and I can trust it	After successful login, the volunteer dashboard and associated data are displayed (FR1.2, 1.3)
	Update my availability status	coordinators know when I am able to help	Availability status is updated immediately in the system (FR5.3)
	View my assigned relief/rescue tasks	I know what work I need to perform	Assigned task list is displayed with task details and location (FR6.3)
	Accept, reject, or update the status of an assigned task	the coordinator is aware of my task progress	Task status is updated and visible to the assigning coordinator (FR6.4)
	Receive notifications about new task assignments and updates	I can respond quickly	Notification is sent when a task is created or updated (FR9.3)
Relief Coordinator	Login with my username and password	the system can authenticate me and I can trust it	After successful login, the coordinator dashboard and associated data are displayed (FR1.2, 1.3)
	View and manage all reported emergencies	I can plan an appropriate disaster response	Emergency list is displayed and can be searched/filtered (FR2.4, FR13.1)

As A/An	I want to	So that	Acceptance Criteria
	Create rescue and relief tasks	field work can be organized and tracked	Task is created with the required details (FR6.1)
	Assign created tasks to suitable available volunteers	relief and rescue work is carried out efficiently	Task is assigned and the volunteer is notified (FR6.2, FR9.3)
	Filter volunteers by skills, location, and availability	I can assign the right person to a task	Filtered volunteer list is displayed based on selected criteria (FR13.3)
	View all submitted emergency and relief requests and update their status	citizens and organizations know the progress	Status changes to Pending, In Progress, Completed, or Rejected, with history maintained (FR11.1, 11.2, 11.4)
	Allocate resources to affected areas or relief requests	relief reaches the people who need it	Distribution is recorded and inventory is updated automatically (FR8.1, 8.3, 8.4)
	Generate reports on emergency requests, relief requests, and resource distribution	I can monitor and evaluate relief operations	Reports are generated for the requested period (FR14.1, 14.2, 14.3)
	View the history of relief activities	I can review past disaster response performance	Historical emergency, request, and distribution records are displayed with date and time (FR16.1, 16.2, 16.3)
Relief Organization / NGO	Register my organization with the platform	I can participate in coordinated relief efforts	Organization profile is created and saved (FR10.1)
	Login with my username and password	the system can authenticate me and I can trust it	After successful login, the organization dashboard and associated data are displayed (FR1.2, 1.3)
	Manage my organization's profile and contact information	other stakeholders can reach and coordinate with us	Profile information is updated and saved successfully (FR10.2)
	Add relief resources to the inventory	available supplies are tracked accurately	Resource type, quantity, and location are recorded (FR7.1, 7.2)

As A/An	I want to	So that	Acceptance Criteria
	Update resource quantities	inventory information stays accurate	Updated quantity is reflected immediately in the resource list (FR7.3)
	View the history of received, allocated, and distributed resources	I can track how our resources were used	Distribution history is displayed for the organization's resources (FR7.4)
	Receive a notification when a resource reaches a low-stock level	I can restock in time	Notification is sent automatically when stock falls below the threshold (FR7.5)
	Manage my organization's relief activities	I can coordinate effectively with authorities and other organizations	Activity records are updated and visible to authorized users (FR10.3)
Administrator	Login with my username and password	the system can authenticate me securely	After successful login, the Admin Dashboard is displayed (FR1.2, 1.3)
	Manage user accounts and assign roles	system access is properly controlled	User information is saved successfully and role-based permissions are applied (FR12.3, FR15.1, 15.3)
	Verify registered volunteers and organizations	only legitimate users can access the system	Verification status changes to Approved or Rejected (FR5.4, FR10.4)
	Restrict users from accessing functions outside their role	the system stays secure for every user type	Users can only access screens/actions permitted for their role (FR15.2)
	View an overview Admin Dashboard of system activity	I can monitor the overall disaster relief operation	Dashboard displays emergency requests, relief requests, volunteers, shelters, and resources (FR12.1, 12.2, 12.4)
	Manage system settings and configuration	the system continues to run correctly for all user roles	Updated system settings are saved and applied across the system (FR15.1, FR15.3)

As A/An	I want to	So that	Acceptance Criteria
	View or download system-wide reports	I can review relief performance for decision-making	Reports are displayed and can be viewed or downloaded (FR14.5)
Management	Login with my username and password	the system can authenticate me and I can trust it	After successful login, the management dashboard and associated data are displayed (FR1.2, 1.3)
	View a system-wide overview of ongoing relief operations	I can understand the overall disaster response situation	Overview displays emergency requests, relief requests, volunteers, shelters, and resources (FR12.1, 12.2, 12.4)
	Review and approve high-level relief and resource requests	resources are released only after proper authorization	Request status is updated to Approved or Rejected and the decision is recorded (FR11.2, 11.4)
	Allocate budget and resources for relief operations	relief organizations and coordinators have what they need to respond effectively	Allocation is recorded and reflected in the resource distribution records (FR8.1, FR14.2)
	View reports on relief operations and resource distribution	I can make informed organizational decisions	Reports are generated and can be viewed or downloaded for the requested period (FR14.1, 14.5)

3.3 Requirements Traceability Matrix

3.3.1 Functional Requirements

1. 1: User Registration and Login

1.1: The system shall allow users such as citizens, volunteers, relief organizations, and administrators to register using required personal information, email/phone number, and password.

1.2: The system shall verify the user's login credentials against the database.

1.3: The system shall redirect users to their respective dashboards based on their roles after successful login.

1.4: The system shall allow users to update their profile information.

Priority Level: High

Precondition: User must provide valid registration information and have a verified account for login.

2. 2: Emergency Reporting

2.1: The system shall allow citizens to report disaster-related emergencies.

2.2: The system shall allow users to provide the emergency type, description, location, and required assistance.

2.3: The system shall record the date and time of each emergency report.

2.4: The system shall allow authorized relief coordinators to view and manage reported emergencies.

Priority Level: High

Precondition: User must be logged in and provide valid emergency information.

3. 3: Relief Assistance Request

3.1: The system shall allow affected citizens to request relief assistance.

3.2: Users shall be able to request different types of assistance such as food, clean water, medicine, clothing, shelter, or rescue.

3.3: The system shall allow users to specify the quantity or level of assistance required.

3.4: The system shall allow users to track the status of their requests.

Priority Level: High

Precondition: User must be registered and logged in.

4. 4: Shelter Management

4.1: The system shall allow authorized users to add and update shelter information.

4.2: The system shall display shelter name, location, capacity, available space, and current status.

4.3: Citizens shall be able to search for available shelters based on their location.

4.4: The system shall allow shelter managers to update the number of available spaces.

Priority Level: High

Precondition: Shelter information must be available in the system, and users must have access to the shelter module.

5. 5: Volunteer Registration

5.1: The system shall allow individuals to register as volunteers.

5.2: Volunteers shall be able to provide their skills, location, contact information, and availability.

5.3: The system shall allow volunteers to update their availability status.

5.4: The system shall allow administrators or coordinators to verify volunteer information.

Priority Level: Medium

Precondition: User must provide valid personal and volunteer information.

6. 6: Volunteer Task Assignment

6.1: The system shall allow relief coordinators to create rescue and relief tasks.

6.2: The system shall allow coordinators to assign tasks to suitable available volunteers.

6.3: Volunteers shall be able to view their assigned tasks.

6.4: Volunteers shall be able to accept, reject, and update the status of assigned tasks.

Priority Level: High

Precondition: Volunteer must be registered and verified, and a task must be available.

7. 7: Relief Resource Management

7.1: The system shall allow authorized organizations to add relief resources to the inventory.

7.2: The system shall record resource type, quantity, location, and availability.

7.3: The system shall allow authorized users to update resource quantities.

7.4: The system shall maintain records of received, allocated, and distributed resources.

7.5: The system shall notify authorized users when important resources reach a low-stock level.

Priority Level: High

Precondition: Authorized organization or administrator must be logged in.

8. 8: Resource Allocation and Distribution

8.1: The system shall allow relief coordinators to allocate resources to affected areas or relief requests.

8.2: The system shall allow users to record the quantity and type of resources distributed.

8.3: The system shall automatically update the available inventory after resource distribution.

8.4: The system shall maintain a history of resource distribution.

Priority Level: High

Precondition: Required resources must be available in the inventory and the user must have authorization to allocate resources.

9. 9: Disaster and Relief Notifications

9.1: The system shall send notifications about important disaster-related information.

9.2: The system shall notify citizens about the status of their relief or emergency requests.

9.3: The system shall notify volunteers about new task assignments and task updates.

9.4: The system shall provide shelter and relief distribution updates to relevant users.

Priority Level: High

Precondition: User must have a registered account and notification service must be available.

10. 10: Organization Management

10.1: The system shall allow NGOs and relief organizations to register with the platform.

10.2: The system shall allow organizations to manage their profiles and contact information.

10.3: Organizations shall be able to manage their available resources and relief activities.

10.4: Authorized administrators shall be able to verify and manage registered organizations.

Priority Level: Medium

Precondition: Organization must provide valid registration information.

11. 11: Emergency and Relief Request Tracking

11.1: The system shall allow coordinators to view all submitted emergency and relief requests.

11.2: The system shall allow authorized users to update request status such as Pending, In Progress, Completed, or Rejected.

11.3: Citizens shall be able to view the current status of their submitted requests.

11.4: The system shall maintain a history of request status changes.

Priority Level: High

Precondition: An emergency or relief request must have been submitted.

12. 12: Admin Dashboard

12.1: The system shall provide administrators with a dashboard containing an overview of system activities.

12.2: The dashboard shall display information about emergency requests, relief requests, volunteers, shelters, and resources.

12.3: Administrators shall be able to manage users and their roles.

12.4: Administrators shall be able to monitor overall disaster relief activities.

Priority Level: High

Precondition: Administrator must be securely logged in with administrative privileges.

13. 13: Search and Filter

13.1: The system shall allow authorized users to search emergency and relief requests.

13.2: Users shall be able to search for available shelters based on location and availability.

13.3: Coordinators shall be able to filter volunteers based on skills, location, and availability.

13.4: Authorized users shall be able to search and filter resources based on type, quantity, and location.

Priority Level: Medium

Precondition: Relevant information must be available in the system.

14. 14: Reports and Monitoring

14.1: The system shall generate reports on emergency and relief requests.

14.2: The system shall generate reports on resource allocation and distribution.

14.3: The system shall generate reports on volunteer activities and assigned tasks.

14.4: The system shall provide shelter occupancy and availability reports.

14.5: Authorized users shall be able to view or download generated reports.

Priority Level: Medium

Precondition: Sufficient data must be available in the system and the user must have reporting privileges.

15. 15: Role-Based Access Control

15.1: The system shall provide different access permissions for citizens, volunteers, relief organizations, shelter managers, coordinators, and administrators.

15.2: The system shall restrict users from accessing functions that are not permitted for their roles.

15.3: The administrator shall be able to manage user roles and permissions.

Priority Level: High

Precondition: User must be authenticated and assigned a valid role.

16. 16: Relief Activity History

16.1: The system shall maintain records of emergency reports, relief requests, resource distribution, volunteer activities, and shelter activities.

16.2: Authorized users shall be able to view historical relief activity records.

16.3: The system shall record the date and time of important relief activities.

Priority Level: Medium

Precondition: Relief activities must have been recorded in the system.

3.3.2 Non-Functional Requirements

17. 1: Usability

1.1: The system shall provide a simple and user-friendly interface that can be used by citizens, volunteers, and relief workers with basic smartphone and computer knowledge.

1.2: The system shall allow users to report an emergency or request relief assistance with minimal steps.

Priority: High

Measurement: User testing should show that at least 90% of users can successfully submit an emergency or relief request without assistance.

18. 2: Availability

2.1: The system shall be available whenever users need to access emergency reporting, relief requests, shelter information, and other critical services.

2.2: The system shall minimize service interruption during disaster situations.

Priority: Very High

Measurement: System uptime monitoring and server availability logs.

19. 3: Security

3.1: The system shall protect user information and authentication credentials from unauthorized access.

3.2: The system shall use secure authentication and authorization mechanisms for different user roles.

3.3: Sensitive information such as user contact details, emergency locations, and organizational data shall only be accessible to authorized users.

Priority: Very High

Measurement: Security testing, access-control testing, and security audit.

20. 4: Performance

4.1: The system shall load important pages such as emergency reporting, relief requests, and shelter information within a reasonable response time under normal network conditions.

4.2: The system shall process emergency and relief requests without significant delay.

4.3: Notifications shall be delivered to intended users as quickly as possible after they are generated.

Priority: High

Measurement: Performance testing and response-time monitoring.

21. 5: Scalability

5.1: The system shall support an increasing number of users, emergency reports, relief requests, volunteers, organizations, and resource records.

5.2: The system should remain functional when a large number of users access the system simultaneously during a major disaster.

Priority: High

Measurement: Load testing and stress testing.

22. 6: Reliability

6.1: The system shall maintain accurate records of emergency reports, relief requests, resources, volunteers, and shelter information.

6.2: The system shall minimize data loss during system failures.

6.3: The system shall provide backup and recovery mechanisms for important system data.

Priority: Very High

Measurement: System reliability testing, backup testing, and recovery testing.

23. 7: Data Integrity

7.1: The system shall ensure that resource quantities, relief distribution records, emergency requests, and other important information remain accurate and consistent.

7.2: The system shall prevent unauthorized users from modifying critical records.

Priority: High

Measurement: Data validation testing and database integrity checks.

24. 8: Maintainability

8.1: The system shall be developed using a structured and modular architecture so that individual components can be maintained or updated easily.

8.2: The system shall allow future features and improvements to be added without significantly affecting existing functionality.

Priority: Medium

Measurement: Code review, documentation review, and maintenance testing.

25. 9: Compatibility

9.1: The system shall work correctly on commonly used web browsers and mobile devices.

9.2: The system shall provide a responsive interface for smartphones, tablets, and computers.

Priority: Medium

Measurement: Cross-browser and cross-device compatibility testing.

26. 10: Localization

10.1: The system shall support both **Bangla and English** languages.

10.2: Users shall be able to select their preferred language through the system interface.

Priority: Medium

Measurement: Language selection testing and verification of translated content.

27. 11: Accessibility

11.1: The system shall present emergency and relief information in a clear and understandable format.

11.2: Important functions such as emergency reporting, shelter information, and relief requests shall be easy to locate and access.

Priority: High

Measurement: Accessibility testing and user evaluation.

28. 12: Backup and Recovery

12.1: The system shall regularly back up important information such as emergency reports, relief requests, resource records, and user data.

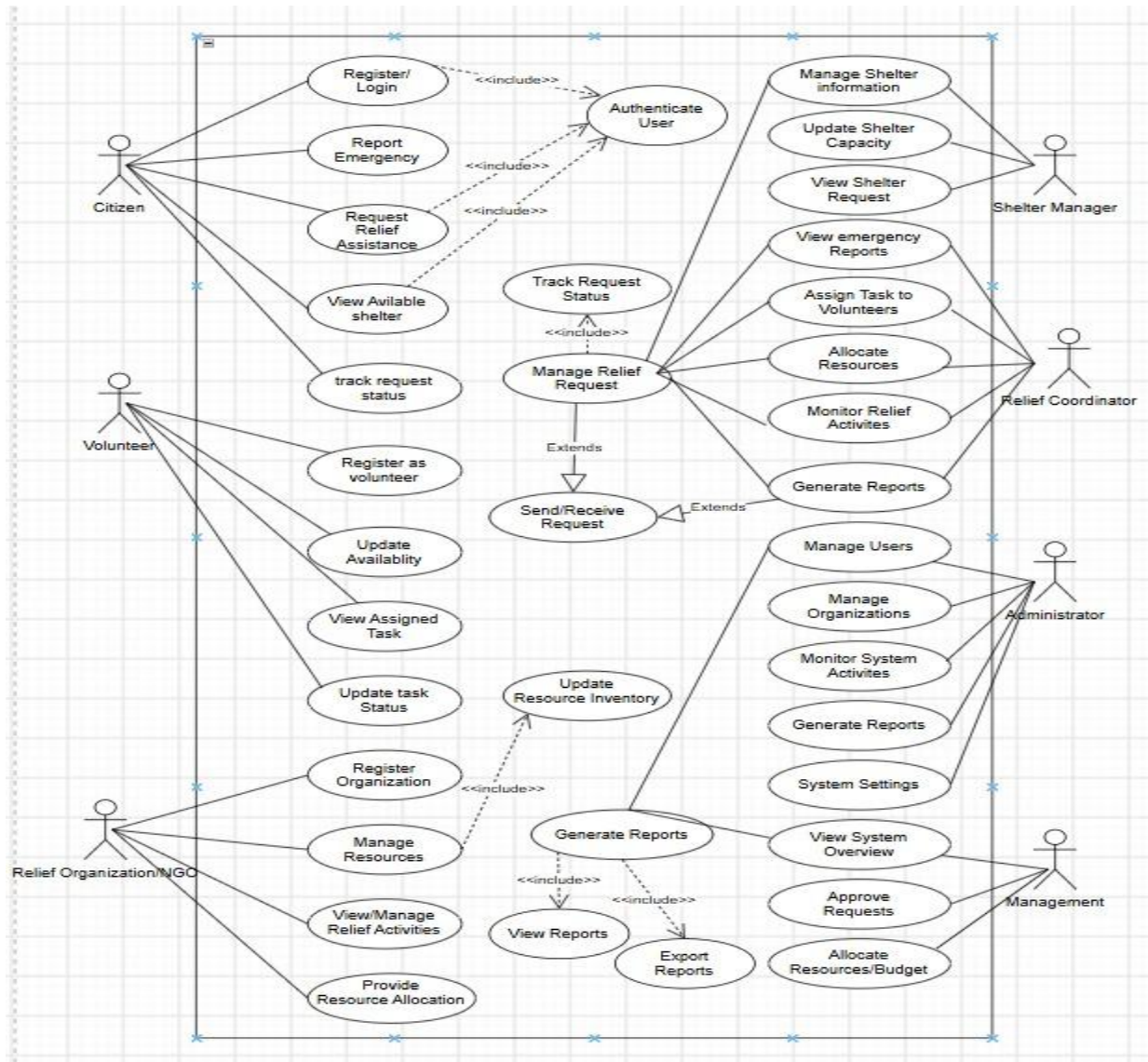
12.2: The system shall provide a recovery mechanism to restore important data after system failure.

Priority: Very High

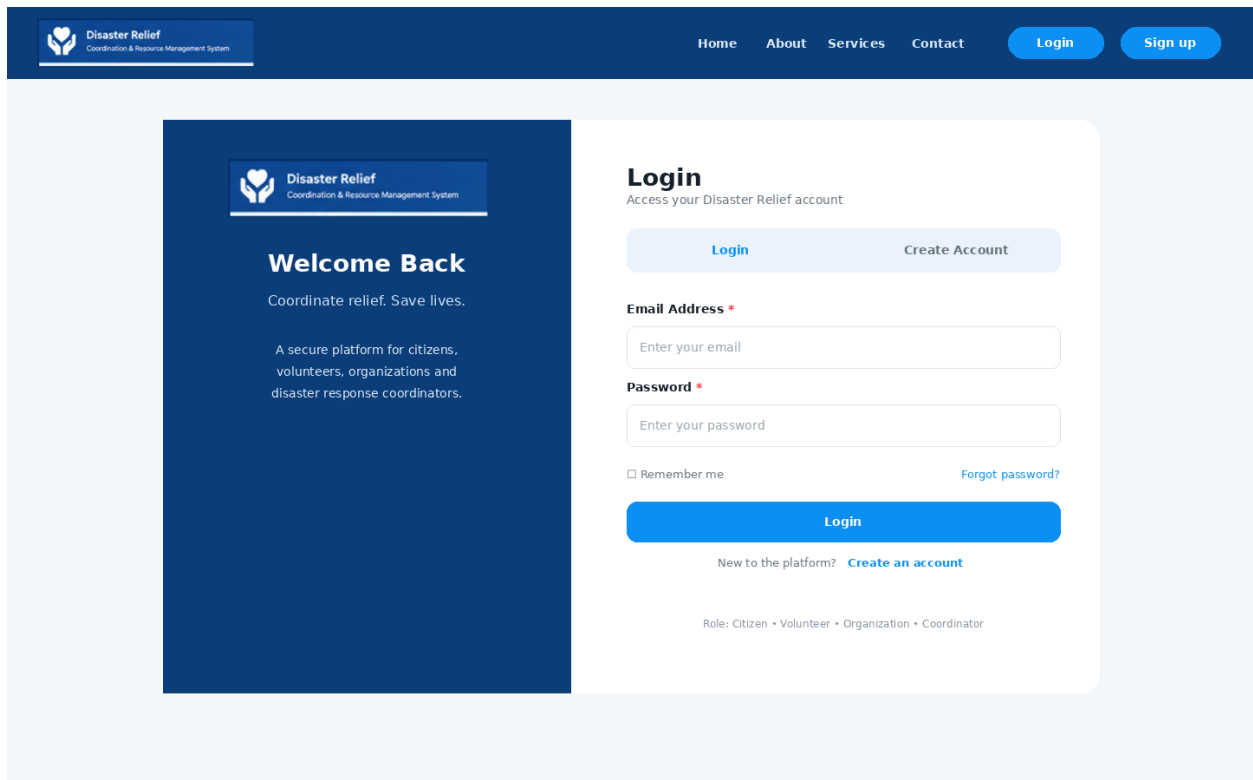
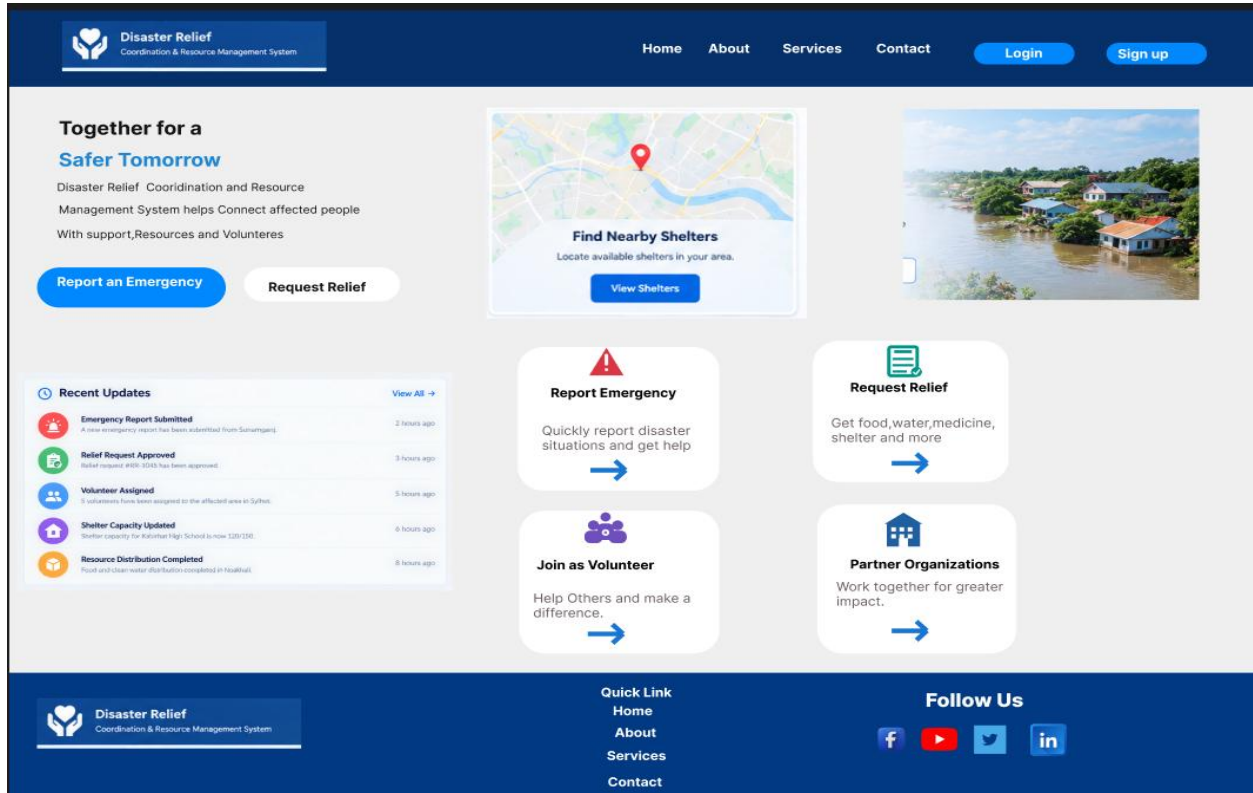
Measurement: Backup and disaster-recovery testing.

4. SOFTWARE DESIGN

4.1 Use Case Diagram



4.2 UI / Wireframe Design using Figma





Disaster Relief
Coordination & Resource Management System

[Home](#)
[About](#)
[Services](#)
[Contact](#)

[Login](#)
[Sign up](#)

Disaster Relief
Coordination & Resource Management System

Dashboard

Emergency Reports

Relief Requests

Shelters

Volunteers

Resources

Notifications

Profile

Citizen Dashboard

Welcome back. Get emergency help and stay informed.

N

Citizen User

Active

3

My Requests

1

Open Emergencies

2

Nearby Shelters

4

Notifications

Report Emergency

Quickly report a disaster situation and request urgent help.

→

Request Relief

Request food, water, medicine, rescue or shelter.

→

Find Shelter

Locate nearby shelters and check capacity.

→

Track Requests

View the status of your emergency and relief requests.

→

Recent Updates

✓

Relief request approved

2 hours ago

✓

Shelter capacity updated

4 hours ago

✓

Volunteer assigned

5 hours ago

✓

Resource distribution completed

8 hours ago

Emergency?

For immediate danger, report the emergency first.

Report Now



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Disaster Relief
Coordination & Resource Management System

Dashboard

Emergency Reports

Relief Requests

Shelters

Volunteers

Resources

Notifications

Profile

Report an Emergency

Provide accurate information so response teams can act quickly.

1

1. Emergency Details

2

2. Location

3

3. Review & Submit

Emergency Type *

Select disaster type

Severity Level *

Select severity

People Affected *

e.g. 10

Location / Address *

Enter affected area or landmark

Contact Number *

Enter reachable phone number

Emergency Description

Describe what happened, immediate needs, and any hazards...

Photo / Evidence



Upload image

Optional

⚠ If there is immediate danger, prioritize evacuation and contact local emergency services.

Submit Emergency



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[Home](#)
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Login

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Disaster Relief

Coordination & Resource Management System

Dashboard

Emergency Reports

Relief Requests

Shelters

Volunteers

Resources

Notifications

Profile

Relief Assistance Request

Tell us what support your household or community needs.

Select assistance needed

Food

Clean Water

Medicine

Shelter

Rescue

Clothing

Quantity / People *

Urgency *

Delivery Location *

e.g. 5 families

Select urgency

Enter address / area

Additional Details

Request status

Add specific needs, medical information, or delivery instructions...

Draft

You can track approval after submission.

Submit Relief Request



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[Home](#)
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[Services](#)
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Coordination & Resource Management System

Dashboard

Emergency Reports

Relief Requests

Shelters

Volunteers

Resources

Notifications

Profile

Shelter Locator

Find nearby shelters and check their current capacity.

Search by area, landmark, or shelter name

Search Shelters

Available Shelters

Nearby shelter map

Community High School

1.2 km

120 / 200 occupied

View details →

Central Cyclone Shelter

2.4 km

185 / 200 occupied

View details →

Riverbank Relief Centre

3.1 km

65 / 150 occupied

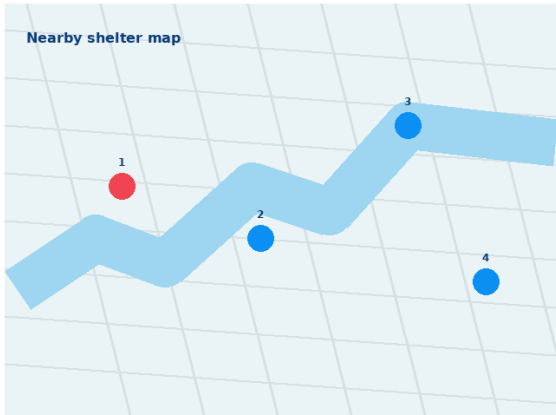
View details →

Municipal Safe House

4.0 km

Full

View details →



Disaster Relief

Coordination & Resource Management System

Home

About

Services

Contact

Login

Sign up

Disaster Relief

Coordination & Resource Management System

Dashboard

Emergency Reports

Relief Requests

Shelters

Volunteers

Resources

Notifications

Profile

Shelter Locator

Find nearby shelters and check their current capacity.

Search by area, landmark, or shelter name

Search Shelters

Available Shelters

Community High School

1.2 km

120 / 200 occupied

View details →

Central Cyclone Shelter

2.4 km

185 / 200 occupied

View details →

Riverbank Relief Center

3.1 km

65 / 150 occupied

View details →

Municipal Safe House

4.0 km

Full

View details →

Nearby shelter map

1

2

3

4

Disaster Relief

Coordination & Resource Management System

Home

About

Services

Contact

Login

Sign up

Disaster Relief

Coordination & Resource Management System

Dashboard

Emergency Reports

Relief Requests

Shelters

Volunteers

Resources

Notifications

Profile

Volunteer Dashboard

Manage your availability, tasks, and relief activities.

Availability

Available

Location: Sunamganj • Skills: First Aid, Distribution

Update Availability

4

Assigned Tasks

2

In Progress

12

Completed

1

Urgent Task

Assigned Tasks

Task	Area	Date	Status	Action
Distribute drinking water	Ward 4	Today, 10:00 AM	In Progress	Open
Assist shelter registration	School Shelter	Today, 2:00 PM	Assigned	Open
Medicine delivery	Riverbank Area	Tomorrow	Assigned	Open
Food package distribution	Village 7	Tomorrow	Completed	View

Page 29 of 52

Disaster Relief

Coordination & Resource
Management System

 [Dashboard](#)

 [Emergency Reports](#)

 [Relief Requests](#)

 [Shelters](#)

 [Volunteers](#)

 [Resources](#)

 [Notifications](#)

 [Profile](#)

Admin / Coordinator Dashboard

Monitor disaster response, people, shelters, and relief resources.

 **24**
Emergency Reports

 **38**
Relief Requests

 **126**
Active Volunteers

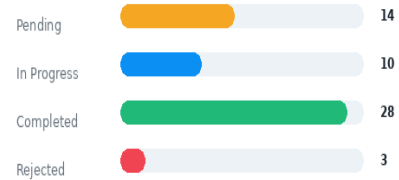
 **7**
Shelters Active

Emergency Reports

Last 7 days



Relief Request Status



Recent Activity

-  Emergency report assigned to Response Team A 10 min ago
-  Food allocation approved for Ward 4 25 min ago
-  Volunteer task completed at Central Shelter 1 hr ago
-  Shelter capacity updated to 185 / 200 2 hrs ago

Quick Actions

-  [Review Emergencies](#)
-  [Approve Relief Requests](#)
-  [Manage Shelters](#)
-  [Generate Report](#)

5. GIT WORKFLOW

https://github.com/nazmusshakib99/Disaster_Relief_Coordination_And_Resource_Management_System-Software_Engineering_Project-.git

The screenshot displays the GitHub web interface for the repository `disaster-relief-coordination-system`. The left sidebar shows the file structure, with the `tasks/nazmus` directory selected. The main area shows a commit history table for the `tasks/nazmus` branch, listing commits from Card 1 to Card 8. Below the table, a terminal window shows the execution of `git` commands to create and commit task files for each card.

Name	Last commit message	Last commit date
..		
card1_resource_inventory_management.txt	[Card 1] Add resource inventory management task file (Nazmus)	20 minutes ago
card2_resource_allocation_distribution.txt	[Card 2] Add resource allocation & distribution task file (Nazmus)	15 minutes ago
card3_notification_system.txt	[Card 3] Add notification system task file (Nazmus)	13 minutes ago
card4_admin_dashboard.txt	[Card 4] Add admin dashboard task file (Nazmus)	12 minutes ago
card5_reports_monitoring.txt	[Card 5] Add reports & monitoring task file (Nazmus)	12 minutes ago
card6_role_based_access_control.txt	[Card 6] Add role-based access control task file (Nazmus)	11 minutes ago
card7_system_integration.txt	[Card 7] Add system integration task file (Nazmus)	11 minutes ago
card8_testing_tc08_tc10.txt	[Card 8] Add testing task file TC_08 to TC_10 (Nazmus)	10 minutes ago

```
ASUSLAPTOP-Q8Q9Q8Q8 MINGW64 ~/Desktop/SE Github Project/jinia/disaster-relief-coordination-system (feature/jinia-all-tasks)
$ mkdir -p tasks/nazmus

ASUSLAPTOP-Q8Q9Q8Q8 MINGW64 ~/Desktop/SE Github Project/jinia/disaster-relief-coordination-system (feature/jinia-all-tasks)
$ cat > tasks/nazmus/card1_resource_inventory_management.txt << 'EOF'
> Card: Card 1
> Title: Resource Inventory Management
> Member: Nazmus Shakib
> Status: In Progress
> Description:
> - Add resource
> - Update quantity
> - Track resource location
> - Track available resources
> - Low-stock notification
> - Test inventory functionality
> EOF
$ git add tasks/nazmus/card1_resource_inventory_management.txt
$ git commit -m "[Card 1] Add resource inventory management task file (Nazmus)"
[feature/jinia-all-tasks 1abc123] [Card 1] Add resource inventory management task file (Nazmus)
1 file changed, 11 insertions(+)
create mode 100644 tasks/nazmus/card1_resource_inventory_management.txt

ASUSLAPTOP-Q8Q9Q8Q8 MINGW64 ~/Desktop/SE Github Project/jinia/disaster-relief-coordination-system (feature/jinia-all-tasks)
$ cat > tasks/nazmus/card2_resource_allocation_distribution.txt << 'EOF'
> Card: Card 2
> Title: Resource Allocation & Distribution
> Member: Nazmus Shakib
> Status: In Progress
> Description:
> - Select relief request
> - Select available resource
> - Allocate quantity
> - Update inventory automatically
> - Store distribution history
> EOF
$ git add tasks/nazmus/card2_resource_allocation_distribution.txt
$ git commit -m "[Card 2] Add resource allocation & distribution task file (Nazmus)"
[feature/jinia-all-tasks 2def456] [Card 2] Add resource allocation & distribution task file (Nazmus)
1 file changed, 9 insertions(+)
create mode 100644 tasks/nazmus/card2_resource_allocation_distribution.txt

ASUSLAPTOP-Q8Q9Q8Q8 MINGW64 ~/Desktop/SE Github Project/jinia/disaster-relief-coordination-system (feature/jinia-all-tasks)
$ cat > tasks/nazmus/card3_notification_system.txt << 'EOF'
> Card: Card 3
> Title: Notification System
> Member: Nazmus Shakib
> Status: In Progress
> Description:
> - Request status notification
> - Volunteer task notification
> - Shelter/relief updates
> - Notification history
> EOF
$ git add tasks/nazmus/card3_notification_system.txt
$ git commit -m "[Card 3] Add notification system task file (Nazmus)"
[feature/jinia-all-tasks 3ghi789] [Card 3] Add notification system task file (Nazmus)
1 file changed, 8 insertions(+)
create mode 100644 tasks/nazmus/card3_notification_system.txt

ASUSLAPTOP-Q8Q9Q8Q8 MINGW64 ~/Desktop/SE Github Project/jinia/disaster-relief-coordination-system (feature/jinia-all-tasks)
$ cat > tasks/nazmus/card4_admin_dashboard.txt << 'EOF'
> Card: Card 4
> Title: Admin Dashboard
> Member: Nazmus Shakib
> Status: In Progress
> Description:
> - User overview
> - Emergency overview
> - Relief request overview
> - Volunteer overview
> - Shelter overview
> - Resource overview
> - UI Design: Admin Dashboard screen
> EOF
$ git add tasks/nazmus/card4_admin_dashboard.txt
$ git commit -m "[Card 4] Add admin dashboard task file (Nazmus)"
[feature/jinia-all-tasks 4jkl012] [Card 4] Add admin dashboard task file (Nazmus)
1 file changed, 12 insertions(+)
create mode 100644 tasks/nazmus/card4_admin_dashboard.txt

ASUSLAPTOP-Q8Q9Q8Q8 MINGW64 ~/Desktop/SE Github Project/jinia/disaster-relief-coordination-system (feature/jinia-all-tasks)
$ cat > tasks/nazmus/card5_reports_monitoring.txt << 'EOF'
> Card: Card 5
> Title: Reports & Monitoring
> Member: Nazmus Shakib
> Status: In Progress
> Description:
> - Emergency reports
> - Relief distribution reports
> - Volunteer activity
> - Shelter occupancy
> EOF
$ git add tasks/nazmus/card5_reports_monitoring.txt
$ git commit -m "[Card 5] Add reports & monitoring task file (Nazmus)"
[feature/jinia-all-tasks 5mno345] [Card 5] Add reports & monitoring task file (Nazmus)
1 file changed, 8 insertions(+)
create mode 100644 tasks/nazmus/card5_reports_monitoring.txt

ASUSLAPTOP-Q8Q9Q8Q8 MINGW64 ~/Desktop/SE Github Project/jinia/disaster-relief-coordination-system (feature/jinia-all-tasks)
$ cat > tasks/nazmus/card6_role_based_access_control.txt << 'EOF'
> Card: Card 6
> Title: Role-Based Access Control
> Member: Nazmus Shakib
> Status: In Progress
> Description:
> - Citizen
> - Volunteer
> - Organization
> - Shelter Manager
> - Coordinator
> - Admin
> EOF
$ git add tasks/nazmus/card6_role_based_access_control.txt
$ git commit -m "[Card 6] Add role-based access control task file (Nazmus)"
[feature/jinia-all-tasks 6pqr678] [Card 6] Add role-based access control task file (Nazmus)
1 file changed, 10 insertions(+)
create mode 100644 tasks/nazmus/card6_role_based_access_control.txt

ASUSLAPTOP-Q8Q9Q8Q8 MINGW64 ~/Desktop/SE Github Project/jinia/disaster-relief-coordination-system (feature/jinia-all-tasks)
$ cat > tasks/nazmus/card7_system_integration.txt << 'EOF'
> Card: Card 7
> Title: System Integration
> Member: Nazmus Shakib
> Status: To Do
> Description:
> - Integrate team members' modules
> - Test database connections
> - Fix integration problems
> - Final system testing
> EOF
$ git add tasks/nazmus/card7_system_integration.txt
$ git commit -m "[Card 7] Add system integration task file (Nazmus)"
[feature/jinia-all-tasks 7stuv901] [Card 7] Add system integration task file (Nazmus)
1 file changed, 8 insertions(+)
create mode 100644 tasks/nazmus/card7_system_integration.txt

ASUSLAPTOP-Q8Q9Q8Q8 MINGW64 ~/Desktop/SE Github Project/jinia/disaster-relief-coordination-system (feature/jinia-all-tasks)
$ cat > tasks/nazmus/card8_testing_tc08_tc10.txt << 'EOF'
> Card: Card 8
> Title: Testing - TC_08 to TC_10
> Member: Nazmus Shakib
> Status: In Progress
> Description:
> - TC_08: Verify organization can add a resource to the inventory.
> - TC_09: Verify coordinator can allocate resources to an affected area
> - TC_10: Verify citizen receives a notification when request status changes
> EOF
$ git add tasks/nazmus/card8_testing_tc08_tc10.txt
$ git commit -m "[Card 8] Add testing task file TC_08 to TC_10 (Nazmus)"
[feature/jinia-all-tasks 8vwxyz234] [Card 8] Add testing task file TC_08 to TC_10 (Nazmus)
1 file changed, 8 insertions(+)
create mode 100644 tasks/nazmus/card8_testing_tc08_tc10.txt
```

CodeIssuesPull requests1ActionsProjectsWikiSecurity and qualityInsightsSettings

Files

feature/Ria-all-...

Go to file

tasks/ria

README.md

disaster-relief-coordination-system / tasks / ria /

atikarahmanria [Card 6] Add testing task file TC_01 to TC_04 (Ria) cbe9ab5 · 23 minutes ago History

This branch is 6 commits ahead of main .

Contribute

Name	Last commit message	Last commit date
..		
card1_user_registration_login.txt	[Card 1] Add user registration & login task file (Ria)	30 minutes ago
card2_emergency_reporting.txt	[Card 2] Add emergency reporting task file (Ria)	28 minutes ago
card3_relief_assistance_request.txt	[Card 3] Add relief assistance request task file (Ria)	27 minutes ago
card4_citizen_dashboard.txt	[Card 4] Add citizen dashboard task file (Ria)	26 minutes ago
card5_ui_design.txt	[Card 5] Add UI design task file (Ria)	25 minutes ago
card6_testing_tc01_tc04.txt	[Card 6] Add testing task file TC_01 to TC_04 (Ria)	23 minutes ago

```
$ git commit -m "[Card 4] Add citizen dashboard task file (Ria)"
[feature/Ria-all-tasks 7e81e77] [Card 4] Add citizen dashboard task file (Ria)
1 file changed, 10 insertions(+)
create mode 100644 tasks/ria/card4_citizen_dashboard.txt

Asus@LAPTOP-ORQONQEO MINGW64 ~/Desktop/SE Github Project/Ria/disaster-relief-coordination-system (feature/Ria-all-tasks)
$ cat > tasks/ria/card5_ui_design.txt << 'EOF'
Card: Card 5
Title: UI Design
Member: Ria Akter
Status: In Progress

Description:
- Login/Registration
- Emergency Reporting
- Relief Assistance Request
EOF

Asus@LAPTOP-ORQONQEO MINGW64 ~/Desktop/SE Github Project/Ria/disaster-relief-coordination-system (feature/Ria-all-tasks)
$ git add tasks/ria/card5_ui_design.txt
warning: in the working copy of 'tasks/ria/card5_ui_design.txt', LF will be replaced by CRLF the next time Git touches it

Asus@LAPTOP-ORQONQEO MINGW64 ~/Desktop/SE Github Project/Ria/disaster-relief-coordination-system (feature/Ria-all-tasks)
$ git commit -m "[Card 5] Add UI design task file (Ria)"
[feature/Ria-all-tasks abc2a6c] [Card 5] Add UI design task file (Ria)
1 file changed, 9 insertions(+)
create mode 100644 tasks/ria/card5_ui_design.txt

Asus@LAPTOP-ORQONQEO MINGW64 ~/Desktop/SE Github Project/Ria/disaster-relief-coordination-system (feature/Ria-all-tasks)
$ cat > tasks/ria/card6_testing_tc01_tc04.txt << 'EOF'
Card: Card 6
Title: Testing - TC_01 to TC_04
Member: Ria Akter
Status: In Progress

Description:
- TC_01: Verify citizen can register with valid details
- TC_02: Verify citizen can log in with valid credentials
- TC_03: Verify citizen can submit an emergency report
- TC_04: Verify citizen can submit a relief assistance request
EOF

Asus@LAPTOP-ORQONQEO MINGW64 ~/Desktop/SE Github Project/Ria/disaster-relief-coordination-system (feature/Ria-all-tasks)
$ git add tasks/ria/card6_testing_tc01_tc04.txt
warning: in the working copy of 'tasks/ria/card6_testing_tc01_tc04.txt', LF will be replaced by CRLF the next time Git touches it

Asus@LAPTOP-ORQONQEO MINGW64 ~/Desktop/SE Github Project/Ria/disaster-relief-coordination-system (feature/Ria-all-tasks)
$ git commit -m "[Card 6] Add testing task file TC_01 to TC_04 (Ria)"
[feature/Ria-all-tasks cbe9ab5] [Card 6] Add testing task file TC_01 to TC_04 (Ria)
1 file changed, 10 insertions(+)
create mode 100644 tasks/ria/card6_testing_tc01_tc04.txt

Asus@LAPTOP-ORQONQEO MINGW64 ~/Desktop/SE Github Project/Ria/disaster-relief-coordination-system (feature/Ria-all-tasks)
$ git push origin feature/Ria-all-tasks
Enumerating objects: 31, done.
Counting objects: 100% (31/31), done.
Delta compression using up to 8 threads
Compressing objects: 100% (24/24), done.
Writing objects: 100% (30/30), 3.05 KiB | 391.00 KiB/s, done.
Total 30 (delta 5), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (5/5), done.
remote:
remote: Create a pull request for 'feature/Ria-all-tasks' on GitHub by visiting:
remote:   https://github.com/nazmusshakib99/disaster-relief-coordination-system/pull/new/feature/Ria-all-tasks
remote:
To https://github.com/nazmusshakib99/disaster-relief-coordination-system.git
 * [new branch]      feature/Ria-all-tasks -> feature/Ria-all-tasks

Asus@LAPTOP-ORQONQEO MINGW64 ~/Desktop/SE Github Project/Ria/disaster-relief-coordination-system (feature/Ria-all-tasks)
$ |
```

Page 32 of 52

nazmusshakib99 / disaster-relief-coordination-system

Code

Issues

Pull requests

Actions

Projects

Wiki

Security and quality

Insights

Settings

Files

disaster-relief-coordination-system / tasks / jinia /

feature/jinia-all...

+

Q

Go to file

tasks/jinia

README.md

disaster-relief-coordination-system / tasks / jinia /

Jiniahossen

[Card 7] Add testing task file TC_05 to TC_07 (Jinia)

17553bf · 1 hour ago

History

This branch is 7 commits ahead of main.

Name

Last commit message

Last commit date

..

card1_system_architecture_design.txt

[Card 1] Add system architecture & design task file (Jinia)

1 hour ago

card2_shelter_management.txt

[Card 2] Add shelter management task file (Jinia)

1 hour ago

card3_shelter_locator.txt

[Card 3] Add shelter locator task file (Jinia)

1 hour ago

card4_volunteer_registration.txt

[Card 4] Add volunteer registration task file (Jinia)

1 hour ago

card5_volunteer_task_assignment.txt

[Card 5] Add volunteer task assignment task file (Jinia)

1 hour ago

card6_ui_design.txt

[Card 6] Add UI design task file (Jinia)

1 hour ago

card7_testing_tc05_tc07.txt

[Card 7] Add testing task file TC_05 to TC_07 (Jinia)

1 hour ago

```
ASUSLAPTOP-08Q0N9ED MINGW64 ~/Desktop/SE Github Project/Jinia/disaster-relief-coordination-system (feature/jinia-all-tasks)
$ git config --global user.name "Jiniahossen"
$ git config --global user.email "jiniahossen1@gmail.com"
$ mkdir -p tasks/jinia
$ cd tasks/jinia
$ echo "Card 1\nTitle: System Architecture & Design\nMember: Jinia Akter Hira\nStatus: In Progress\n\nDescription:\n - Design the overall system architecture\n - Define major modules and components\n - Create system diagrams" > card1_system_architecture_design.txt
$ git add card1_system_architecture_design.txt
$ git commit -m "Card 1 Added: System Architecture & Design (Jinia)"
[feature/jinia-all-tasks 1a2b3c4] Card 1 Added: System Architecture & Design (Jinia)
1 file changed, 9 insertions(+)
create mode 100644 tasks/jinia/card1_system_architecture_design.txt
ASUSLAPTOP-08Q0N9ED MINGW64 ~/Desktop/SE Github Project/Jinia/disaster-relief-coordination-system (feature/jinia-all-tasks)
$ echo "Card 2\nTitle: Shelter Management\nMember: Jinia Akter Hira\nStatus: In Progress\n\nDescription:\n - Manage shelter details\n - Add, update and delete shelters\n - View shelter capacity and availability" > card2_shelter_management.txt
$ git add card2_shelter_management.txt
$ git commit -m "Card 2 Added: Shelter Management (Jinia)"
[feature/jinia-all-tasks 2b3c4d5] Card 2 Added: Shelter Management (Jinia)
1 file changed, 9 insertions(+)
create mode 100644 tasks/jinia/card2_shelter_management.txt
ASUSLAPTOP-08Q0N9ED MINGW64 ~/Desktop/SE Github Project/Jinia/disaster-relief-coordination-system (feature/jinia-all-tasks)
$ echo "Card 3\nTitle: Shelter Locator\nMember: Jinia Akter Hira\nStatus: In Progress\n\nDescription:\n - Allow users to search nearby shelters\n - Show location on map\n - Filter by facilities and capacity" > card3_shelter_locator.txt
$ git add card3_shelter_locator.txt
$ git commit -m "Card 3 Added: Shelter Locator (Jinia)"
[feature/jinia-all-tasks 3c4d5e6] Card 3 Added: Shelter Locator (Jinia)
1 file changed, 9 insertions(+)
create mode 100644 tasks/jinia/card3_shelter_locator.txt
ASUSLAPTOP-08Q0N9ED MINGW64 ~/Desktop/SE Github Project/Jinia/disaster-relief-coordination-system (feature/jinia-all-tasks)
$ echo "Card 4\nTitle: Volunteer Registration\nMember: Jinia Akter Hira\nStatus: In Progress\n\nDescription:\n - Volunteers can register an account\n - Provide personal and skill details\n - Email verification" > card4_volunteer_registration.txt
$ git add card4_volunteer_registration.txt
$ git commit -m "Card 4 Added: Volunteer Registration (Jinia)"
[feature/jinia-all-tasks 4b5e6f7] Card 4 Added: Volunteer Registration (Jinia)
1 file changed, 9 insertions(+)
create mode 100644 tasks/jinia/card4_volunteer_registration.txt
ASUSLAPTOP-08Q0N9ED MINGW64 ~/Desktop/SE Github Project/Jinia/disaster-relief-coordination-system (feature/jinia-all-tasks)
$ echo "Card 5\nTitle: Volunteer Task Assignment\nMember: Jinia Akter Hira\nStatus: In Progress\n\nDescription:\n - Assign tasks to volunteers\n - Track task status\n - Manage task priorities" > card5_volunteer_task_assignment.txt
$ git add card5_volunteer_task_assignment.txt
$ git commit -m "Card 5 Added: Volunteer Task Assignment (Jinia)"
[feature/jinia-all-tasks 5e6f7g8] Card 5 Added: Volunteer Task Assignment (Jinia)
1 file changed, 9 insertions(+)
create mode 100644 tasks/jinia/card5_volunteer_task_assignment.txt
ASUSLAPTOP-08Q0N9ED MINGW64 ~/Desktop/SE Github Project/Jinia/disaster-relief-coordination-system (feature/jinia-all-tasks)
$ echo "Card 6\nTitle: UI Design\nMember: Jinia Akter Hira\nStatus: In Progress\n\nDescription:\n - Design user interface\n - Create wireframes and mockups\n - Ensure responsive design" > card6_ui_design.txt
$ git add card6_ui_design.txt
$ git commit -m "Card 6 Added: UI Design (Jinia)"
[feature/jinia-all-tasks d67g8h9] Card 6 Added: UI Design (Jinia)
1 file changed, 9 insertions(+)
create mode 100644 tasks/jinia/card6_ui_design.txt
ASUSLAPTOP-08Q0N9ED MINGW64 ~/Desktop/SE Github Project/Jinia/disaster-relief-coordination-system (feature/jinia-all-tasks)
$ echo "Card 7\nTitle: Testing - TC_05 to TC_07\nMember: Jinia Akter Hira\nStatus: In Progress\n\nDescription:\n - TC_05: Verify volunteer registration\n - TC_06: Verify shelter search\n - TC_07: Verify task assignment" > card7_testing_tc05_to_tc07.txt
$ git add card7_testing_tc05_to_tc07.txt
$ git commit -m "Card 7 Added: Testing - TC_05 to TC_07 (Jinia)"
[feature/jinia-all-tasks 9gh9i0j] Card 7 Added: Testing - TC_05 to TC_07 (Jinia)
1 file changed, 9 insertions(+)
create mode 100644 tasks/jinia/card7_testing_tc05_to_tc07.txt
ASUSLAPTOP-08Q0N9ED MINGW64 ~/Desktop/SE Github Project/Jinia/disaster-relief-coordination-system (feature/jinia-all-tasks)
$ git push origin feature/jinia-all-tasks
Enumerating objects: 31, done.
Counting objects: 100% (31/31), done.
Delta compression using up to 8 threads
Compressing objects: 100% (22/22), done.
Writing objects: 100% (31/31), 4.02 KiB | 412.00 KiB/s, done.
Total 31 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (5/5), done.
To https://github.com/nazmusshakib99/disaster-relief-coordination-system.git
 [new branch] feature/jinia-all-tasks -> feature/jinia-all-tasks
ASUSLAPTOP-08Q0N9ED MINGW64 ~/Desktop/SE Github Project/Jinia/disaster-relief-coordination-system (feature/jinia-all-tasks)
$
```

Page 33 of 52

```
user@DESKTOP-4B4F1B9 MINGW64 /d/Ashli GitHub
$ git clone https://github.com/nazmusshah99/disaster-relief-coordination-system.git
Cloning into 'disaster-relief-coordination-system'...
remote: Enumerating objects: 108, done.
remote: Counting objects: 100% (108/108), done.
remote: Compressing objects: 100% (67/67), done.
remote: Total 108 (delta 18), reused 105 (delta 18), pack-reused 0 (from 0)
Receiving objects: 100% (108/108), 11.72 KiB | 666.00 KiB/s, done.
Resolving deltas: 100% (18/18), done.

user@DESKTOP-4B4F1B9 MINGW64 /d/Ashli GitHub
$ cd disaster-relief-coordination-system
$
user@DESKTOP-4B4F1B9 MINGW64 /d/Ashli GitHub/disaster-relief-coordination-system (main)
$
user@DESKTOP-4B4F1B9 MINGW64 /d/Ashli GitHub/disaster-relief-coordination-system (main)
$ git checkout -b feature/ashli-all-tasks
Switched to a new branch 'feature/ashli-all-tasks'

user@DESKTOP-4B4F1B9 MINGW64 /d/Ashli GitHub/disaster-relief-coordination-system (feature/ashli-all-tasks)
$ mkdir -p tasks/ashli

user@DESKTOP-4B4F1B9 MINGW64 /d/Ashli GitHub/disaster-relief-coordination-system (feature/ashli-all-tasks)
$ cat > tasks/ashli/card1_organization_management.txt << 'EOF'
Card: Card 1
Title: Organization Management
Member: Ashli
Status: In Progress

Description:
- NGO registration
- Organization profile
- Contact information
- Organization verification
EOF

user@DESKTOP-4B4F1B9 MINGW64 /d/Ashli GitHub/disaster-relief-coordination-system (feature/ashli-all-tasks)
$ git add tasks/ashli/card1_organization_management.txt
warning: In the working copy of 'tasks/ashli/card1_organization_management.txt', LF will be replaced by CRLF the next time Git touches it

user@DESKTOP-4B4F1B9 MINGW64 /d/Ashli GitHub/disaster-relief-coordination-system (feature/ashli-all-tasks)
$ git commit -c '[Card 1] Add organization management task file (Ashli)'
[feature/ashli-all-tasks f38d6ad] [Card 1] Add organization management task file (Ashli)
1 file changed, 10 insertions(+), 1 deletion(-)
create mode 100644 tasks/ashli/card1_organization_management.txt

user@DESKTOP-4B4F1B9 MINGW64 /d/Ashli GitHub/disaster-relief-coordination-system (feature/ashli-all-tasks)
$ cat > tasks/ashli/card2_organization_resources.txt << 'EOF'
Card: Card 2
Title: Organization Resources
Member: Ashli
Status: In Progress

Description:
- Organization activities
- Resource information
- Relief activities
```

Code

Issues

Pull requests

Agents

Actions

Projects

Wiki

Security and quality

Insights

Settings

Files

feature/ashil-all...

Go to file

tasks/ashil

README.md

disaster-relief-coordination-system / tasks / ashil

samanashil1234567890-sudo

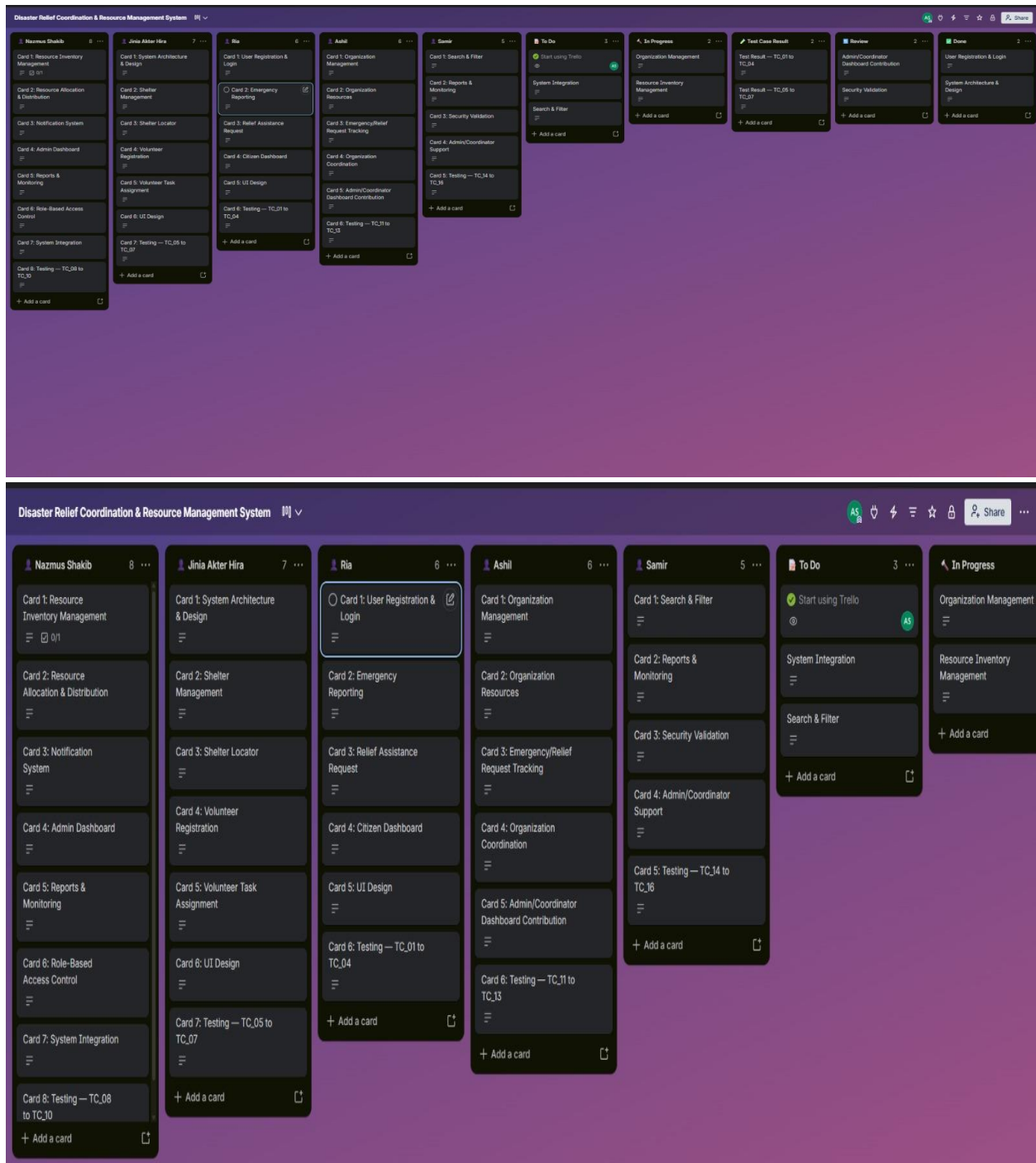
[Card 6] Add testing task file TC_11 to TC_13 (Ashil)

e8a11e0 · 1 minute ago

History

This branch is 4 commits ahead of main.

Name	Last commit message	Last commit date
..		
card1_organization_management.txt	[Card 1] Add organization management task file (Ashil)	6 minutes ago
card4_organization_coordination.txt	[Card 4] Add organization coordination task file (Ashil)	4 minutes ago
card5_admin_coordinator_dashboard_contribution.txt	[Card 5] Add admin/coordinator dashboard contribution task file (Ashil)	2 minutes ago
card6_testing_tc11_tc13.txt	[Card 6] Add testing task file TC_11 to TC_13 (Ashil)	1 minute ago



6. TESTING

These are all the test cases designed for our project

TC_01: Verify user registration with valid information

Project Name: Disaster Relief Coordination and Resource Management System			Test Designed by: Atika Rahman Ria	
Test Case ID: TC_01			Test Designed date: 30/08/2026	
Test Priority (Low, Medium, High): High			Test Executed by: Sadman Rafi	
Module Name: User Registration & Login			Test Execution date: 03/09/2026	
Test Title: Verify user registration with valid information				
Description: Ensure a new user can create an account by providing valid personal details.				
Precondition: User is on the registration page and does not already have an account.				
Dependencies: None				
Test Steps	Test Data	Expected Results	Actual Results	Status (Pass/Fail)

1. Open the registration page	Name: Rafiqul Islam			
2. Enter full name, email, phone number and password	Email: rafiqul.islam@gmail.com	Account is created successfully and the user is redirected to the login page.	As expected	Pass
3. Select account type as Citizen	Phone: 01711223344			
4. Click Register	Password: Rafiq@2026			
Post Condition: New user record is stored in the database and can be used to log in.				

TC_02: Verify login with valid username and password

Project Name: Disaster Relief Coordination and Resource Management System			Test Designed by: Atika Rahman Ria	
Test Case ID: TC_02			Test Designed date: 30/08/2026	
Test Priority (Low, Medium, High): High			Test Executed by: Sadman Rafi	
Module Name: User Registration & Login			Test Execution date: 03/09/2026	
Test Title: Verify login with valid username and password				
Description: Ensure a registered user can log in and reach the correct role-based dashboard.				
Precondition: User already has a verified account.				
Dependencies: TC_01 (a registered account must exist)				
Test Steps	Test Data	Expected Results	Actual Results	Status (Pass/Fail)

1. Go to the login page 2. Enter registered email and password 3. Click Login	Email: rafiquel.islam@gmail.com Password: Rafiq@2026	User is authenticated and redirected to the Citizen dashboard with their data loaded.	As expected	Pass
Post Condition: Session is created and the user's dashboard reflects their account information.				

TC_03: Verify citizen can report a disaster-related emergency

Project Name: Disaster Relief Coordination and Resource Management System			Test Designed by: Atika Rahman Ria	
Test Case ID: TC_03			Test Designed date: 30/08/2026	
Test Priority (Low, Medium, High): High			Test Executed by: Sadman Rafi	
Module Name: Emergency Reporting			Test Execution date: 03/09/2026	
Test Title: Verify citizens can report a disaster-related emergency				
Description: Ensure a citizen can submit an emergency report with all required details.				
Precondition: User is logged in as a Citizen.				
Dependencies: None				
Test Steps	Test Data	Expected Results	Actual Results	Status (Pass/Fail)

1. Navigate to Report Emergency	Type: Flood			
2. Select emergency type as Flood	Location: Sylhet, Zinda bazar	Emergency report is saved with the date and time and becomes visible to relief coordinators.	As expected	Pass
3. Enter location and description	Description: Water entering ground-floor homes			
4. Click Submit Report				
Post Condition: Report record is created and appears in the coordinator's emergency list.				

TC_04: Verify citizens can request relief assistance

Project Name: Disaster Relief Coordination and Resource Management System			Test Designed by: Atika Rahman Ria	
Test Case ID: TC_04			Test Designed date: 30/08/2026	
Test Priority (Low, Medium, High): High			Test Executed by: Sadman Rafi	
Module Name: Relief Assistance Request			Test Execution date: 03/09/2026	
Test Title: Verify citizens can request relief assistance				
Description: Ensure a citizen can request a specific type and quantity of relief assistance.				
Precondition: User is logged in as a Citizen with an active emergency report.				
Dependencies: TC_03 (an emergency report should exist)				
Test Steps	Test Data	Expected Results	Actual Results	Status (Pass/Fail)

1. Go to Request Relief 2. Select assistance type as Food & Water 3. Enter quantity required 4. Click Submit Request	Type: Food & Water Quantity: For 5 family members	Request is saved with status Pending and linked to the citizen's account.	As expected	Pass
Post Condition: Relief request appears in the coordinator's request list with status Pending.				

TC_05: Verify shelter manager can add a new shelter

Project Name: Disaster Relief Coordination and Resource Management System			Test Designed by: Jinia Akter Hira	
Test Case ID: TC_05			Test Designed date: 30/08/2026	
Test Priority (Low, Medium, High): High			Test Executed by: Sadman Rafi	
Module Name: Shelter Management			Test Execution date: 03/09/2026	
Test Title: Verify shelter manager can add a new shelter				
Description: Ensure an authorized shelter manager can register a new shelter in the system.				
Precondition: User is logged in with the Shelter Manager role.				
Dependencies: None				
Test Steps	Test Data	Expected Results	Actual Results	Status (Pass/Fail)

1. Navigate to Add Shelter 2. Enter shelter name, location and capacity 3. Click Save	Name: Dhanmondi Community Shelter Location: Dhanmondi, Dhaka Capacity: 100	Shelter is saved and appears correctly in the shelter list.	As expected	Pass
Post Condition: Shelter record is stored and becomes searchable by citizens.				

TC_06: Verify volunteer registration with skills and availability

Project Name: Disaster Relief Coordination and Resource Management System			Test Designed by: Jinia Akter Hira	
Test Case ID: TC_06			Test Designed date: 30/08/2026	
Test Priority (Low, Medium, High): Medium			Test Executed by: Sadman Rafi	
Module Name: Volunteer Registration			Test Execution date: 03/09/2026	
Test Title: Verify volunteer registration with skills and availability				
Description: Ensure a person can register as a volunteer with skills, location and availability.				
Precondition: User is on the volunteer registration page.				
Dependencies: None				
Test Steps	Test Data	Expected Results	Actual Results	Status (Pass/Fail)

1. Open Volunteer Registration	Skill: First Aid			
2. Enter personal information	Location: Mirpur, Dhaka	Volunteer profile is saved with status Pending Verification.	As expected	Pass
3. Enter skill, location and availability	Availability: Weekends			
4. Click Register				
Post Condition: Volunteer record awaits verification by an administrator or coordinator.				

TC_07: Verify coordinator can assign a task to a verified volunteer

Project Name: Disaster Relief Coordination and Resource Management System			Test Designed by: Jinia Akter Hira	
Test Case ID: TC_07			Test Designed date: 30/08/2026	
Test Priority (Low, Medium, High): High			Test Executed by: Sadman Rafi	
Module Name: Volunteer Task Assignment			Test Execution date: 03/09/2026	
Test Title: Verify coordinator can assign a task to a verified volunteer				
Description: Ensure a relief coordinator can create a task and assign it to a suitable volunteer.				
Precondition: A verified volunteer with a matching skill is available.				
Dependencies: TC_06 (volunteer must be registered and verified)				
Test Steps	Test Data	Expected Results	Actual Results	Status (Pass/Fail)

1. Login as Relief Coordinator 2. Create a relief task 3. Select a verified volunteer 4. Click Assign Task	Task: Deliver food packets to Zindabazar Volunteer: verified volunteer with First Aid skill	Task is assigned and the volunteer receives a notification.	As expected	Pass
Post Condition: Task appears in the volunteer's assigned task list with status Assigned.				

TC_08: Verify organization can add a resource to the inventory

Project Name: Disaster Relief Coordination and Resource Management System			Test Designed by: Nazmus Shakib	
Test Case ID: TC_08			Test Designed date: 30/08/2026	
Test Priority (Low, Medium, High): High			Test Executed by: Sadman Rafi	
Module Name: Relief Resource Management			Test Execution date: 03/09/2026	
Test Title: Verify organization can add a resource to the inventory				
Description: Ensure an authorized organization can record a new relief resource.				
Precondition: User is logged in with the Organization role.				
Dependencies: None				
Test Steps	Test Data	Expected Results	Actual Results	Status (Pass/Fail)

1. Navigate to Resource Inventory 2. Enter resource type, quantity and location 3. Click Save	Type: Rice Quantity: 500 kg Location: Warehouse-1, Mirpur	Resource is added and the updated quantity is reflected in the inventory list.	As expected	Pass
Post Condition: Inventory record is updated and the resource becomes available for allocation.				

TC_09: Verify coordinator can allocate resources to an affected area

Project Name: Disaster Relief Coordination and Resource Management System			Test Designed by: Nazmus Shakib	
Test Case ID: TC_09			Test Designed date: 30/08/2026	
Test Priority (Low, Medium, High): High			Test Executed by: Sadman Rafi	
Module Name: Resource Allocation & Distribution			Test Execution date: 03/09/2026	
Test Title: Verify coordinator can allocate resources to an affected area				
Description: Ensure a coordinator can allocate inventory resources to a relief request.				
Precondition: Required resources are available in the inventory.				
Dependencies: TC_04, TC_08 (a request and available inventory must exist)				
Test Steps	Test Data	Expected Results	Actual Results	Status (Pass/Fail)

1. Login as Relief Coordinator				
2. Select an approved relief request	Resource: Rice, 50 kg	Distribution is recorded and the inventory quantity is updated automatically.	As expected	Pass
3. Enter resource and quantity to allocate	Area: Sylhet, Zindabazar			
4. Click Confirm Distribution				
Post Condition: Inventory count is reduced and the distribution is added to the history log.				

TC_10: Verify citizen receives a notification when request status changes

Project Name: Disaster Relief Coordination and Resource Management System			Test Designed by: Nazmus Shakib	
Test Case ID: TC_10			Test Designed date: 30/08/2026	
Test Priority (Low, Medium, High): High			Test Executed by: Sadman Rafi	
Module Name: Disaster & Relief Notifications			Test Execution date: 03/09/2026	
Test Title: Verify citizen receives a notification when request status changes				
Description: Ensure the system sends a notification when a relief request status is updated.				
Precondition: A relief request has already been submitted by the citizen.				
Dependencies: TC_04				
Test Steps	Test Data	Expected Results	Actual Results	Status (Pass/Fail)

1. Coordinator changes request status from Pending to In Progress 2. Citizen opens the notification panel	Request status change: Pending -> In Progress	A notification is delivered promptly to the citizen's account.	As expected	Pass
Post Condition: Notification is stored in the citizen's notification history.				

TC_11: Verify NGO can register an organization profile

Project Name: Disaster Relief Coordination and Resource Management System			Test Designed by: S.M Ashil	
Test Case ID: TC_11			Test Designed date: 30/08/2026	
Test Priority (Low, Medium, High): Medium			Test Executed by: Sadman Rafi	
Module Name: Organization Management			Test Execution date: 03/09/2026	
Test Title: Verify NGO can register an organization profile				
Description: Ensure an NGO can create and save an organization profile on the platform.				
Precondition: User is on the organization registration page.				
Dependencies: None				
Test Steps	Test Data	Expected Results	Actual Results	Status (Pass/Fail)

1. Open Organization Registration	Organization: Hope Foundation	Organization profile is created and saved successfully.	As expected	Pass
2. Enter organization name and contact information	Contact: contact@hopefoundation.org			
3. Click Register				
Post Condition: Organization record awaits admin verification before receiving full access.				

TC_12: Verify coordinator can update the status of a submitted request

Project Name: Disaster Relief Coordination and Resource Management System			Test Designed by: S.M Ashil	
Test Case ID: TC_12			Test Designed date: 30/08/2026	
Test Priority (Low, Medium, High): High			Test Executed by: Sadman Rafi	
Module Name: Emergency & Relief Request Tracking			Test Execution date: 03/09/2026	
Test Title: Verify coordinator can update the status of a submitted request				
Description: Ensure an authorized coordinator can change a request's status and that history is kept.				
Precondition: A relief request has already been submitted.				
Dependencies: TC_04				
Test Steps	Test Data	Expected Results	Actual Results	Status (Pass/Fail)

1. Login as Relief Coordinator				
2. Open a submitted request	Request ID: REQ_101	Status is updated, visible to the citizen, and the change is recorded in the history.	As expected	Pass
3. Change status to In Progress	New status: In Progress			
4. Click Save				
Post Condition: Status-change entry is logged in the request's history.				

TC_13: Verify admin dashboard displays a system-wide overview

Project Name: Disaster Relief Coordination and Resource Management System			Test Designed by: S.M Ashil	
Test Case ID: TC_13			Test Designed date: 30/08/2026	
Test Priority (Low, Medium, High): High			Test Executed by: Sadman Rafi	
Module Name: Admin Dashboard			Test Execution date: 03/09/2026	
Test Title: Verify admin dashboard displays a system-wide overview				
Description: Ensure the Admin Dashboard shows a summary of key system activity.				
Precondition: User is logged in as Administrator.				
Dependencies: None				
Test Steps	Test Data	Expected Results	Actual Results	Status (Pass/Fail)

1. Login as Administrator 2. Navigate to the Admin Dashboard	N/A	Dashboard displays emergency requests, relief requests, volunteers, shelters and resources.	As expected	Pass
Post Condition: Administrator can view a real-time summary of overall relief activity.				

TC_14: Verify coordinator can filter volunteers by skill and location

Project Name: Disaster Relief Coordination and Resource Management System			Test Designed by: Samir Ahmed	
Test Case ID: TC_14			Test Designed date: 30/08/2026	
Test Priority (Low, Medium, High): Medium			Test Executed by: Sadman Rafi	
Module Name: Search and Filter			Test Execution date: 03/09/2026	
Test Title: Verify coordinator can filter volunteers by skill and location				
Description: Ensure the system returns volunteers matching selected filter criteria.				
Precondition: Multiple volunteers with different skills and locations are registered.				
Dependencies: TC_06				
Test Steps	Test Data	Expected Results	Actual Results	Status (Pass/Fail)
1. Login as Relief Coordinator 2. Open the Volunteer List 3. Apply filter: skill = First Aid, location = Mirpur	Skill: First Aid Location: Mirpur	Filtered volunteer list is displayed matching the selected criteria.	As expected	Pass

Post Condition: Filtered results remain available for task assignment.

TC_15: Verify authorized user can generate a resource distribution report

Project Name: Disaster Relief Coordination and Resource Management System			Test Designed by: Samir Ahmed	
Test Case ID: TC_15			Test Designed date: 30/08/2026	
Test Priority (Low, Medium, High): Medium			Test Executed by: Sadman Rafi	
Module Name: Reports and Monitoring			Test Execution date: 03/09/2026	
Test Title: Verify authorized user can generate a resource distribution report				
Description: Ensure a coordinator or admin can generate a report for a selected period.				
Precondition: Sufficient distribution data exists for the selected period.				
Dependencies: TC_09				
Test Steps	Test Data	Expected Results	Actual Results	Status (Pass/Fail)
1. Login as Coordinator/Administrator 2. Navigate to Reports 3. Select report type and date range 4. Click Generate	Report type: Resource Distribution Period: August 2026	Report is generated and made available for viewing or download.	As expected	Pass
Post Condition: Generated report is saved and accessible for future reference.				

TC_16: Verify an unauthorized role cannot access restricted admin functions

Project Name: Disaster Relief Coordination and Resource Management System			Test Designed by: Samir Ahmed	
Test Case ID: TC_16			Test Designed date: 30/08/2026	
Test Priority (Low, Medium, High): High			Test Executed by: Sadman Rafi	
Module Name: Role-Based Access Control			Test Execution date: 03/09/2026	
Test Title: Verify an unauthorized role cannot access restricted admin functions				
Description: Ensure users can only access features permitted for their assigned role.				
Precondition: User is logged in with the Volunteer role.				
Dependencies: None				
Test Steps	Test Data	Expected Results	Actual Results	Status (Pass/Fail)
1. Login as Volunteer 2. Attempt to open the Admin Dashboard via URL or menu	Role: Volunteer	Access is denied and the user is redirected with a restricted-access message.	As expected	Pass
Post Condition: The unauthorized access attempt is logged by the system.				

7. CONCLUSION

The **Disaster Relief Coordination and Resource Management System** is proposed as a digital solution to improve the efficiency and coordination of disaster relief activities in Bangladesh. Natural disasters can create serious challenges for affected communities, relief organizations, volunteers, and government authorities, particularly when communication and resource management are handled through disconnected or manual processes. The proposed system aims to address these challenges by providing a centralized platform for managing emergency reports, relief requests, shelters, volunteers, resources, notifications, and relief activities.

The **Scrum model** has been selected as the software development process because it supports incremental development, continuous testing, regular feedback, and flexible requirements. The system can be developed through multiple Sprints, with each Sprint focusing on specific features and producing a working system increment. Functional and non-functional requirements have been identified to ensure that the system provides the required services while maintaining usability, security, reliability, performance, availability, and scalability.

Overall, the proposed system can help improve communication between stakeholders, reduce delays in relief operations, support better resource allocation, and provide affected people with timely information and assistance. Therefore, the system has the potential to contribute to more organized, efficient, and responsive disaster relief management.